MSG No. 87.79 $I4'/m'$ [Type III, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `4'/m'..`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,4,5,6,7,8]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[9,10,11,12,13,14,15,16]`

Table 2: Wyckoff site: `2b`, site symmetry: `4'/m'..`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[9,10,11,12,13,14,15,16]`

Table 3: Wyckoff site: `4c`, site symmetry: `2/m'..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,2,7,8]`
2	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[3,4,5,6]`
3	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[9,10,15,16]`
4	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[11,12,13,14]`

Table 4: Wyckoff site: `4d`, site symmetry: `-4..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,2,11,12]`
2	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[3,4,9,10]`
3	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[5,6,15,16]`
4	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[7,8,13,14]`

Table 5: Wyckoff site: `4e`, site symmetry: `4'..`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,5,6]`
2	$[0,\ 0,\ -z]$	`[3,4,7,8]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,10,13,14]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[11,12,15,16]`

Table 6: Wyckoff site: `8f`, site symmetry: `-1'`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,15]`
2	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[2,16]`
3	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[3,13]`

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[4,14]
5	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[5,11]
6	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[6,12]
7	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[7,9]
8	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[8,10]

Table 7: Wyckoff site: 8g, site symmetry: 2..

No.	position	mapping
1	$\left[0, \frac{1}{2}, z\right]$	[1,2]
2	$\left[\frac{1}{2}, 0, -z\right]$	[3,4]
3	$\left[\frac{1}{2}, 0, z\right]$	[5,6]
4	$\left[0, \frac{1}{2}, -z\right]$	[7,8]
5	$\left[\frac{1}{2}, 0, z+\frac{1}{2}\right]$	[9,10]
6	$\left[0, \frac{1}{2}, \frac{1}{2}-z\right]$	[11,12]
7	$\left[0, \frac{1}{2}, z+\frac{1}{2}\right]$	[13,14]
8	$\left[\frac{1}{2}, 0, \frac{1}{2}-z\right]$	[15,16]

Table 8: Wyckoff site: 8h, site symmetry: m'..

No.	position	mapping
1	$[x, y, 0]$	[1,8]
2	$[-x, -y, 0]$	[2,7]
3	$[y, -x, 0]$	[3,6]
4	$[-y, x, 0]$	[4,5]
5	$\left[x+\frac{1}{2}, y+\frac{1}{2}, \frac{1}{2}\right]$	[9,16]
6	$\left[\frac{1}{2}-x, \frac{1}{2}-y, \frac{1}{2}\right]$	[10,15]
7	$\left[y+\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$	[11,14]
8	$\left[\frac{1}{2}-y, x+\frac{1}{2}, \frac{1}{2}\right]$	[12,13]

Table 9: Wyckoff site: 16i, site symmetry: 1

No.	position	mapping
1	$[x, y, z]$	[1]
2	$[-x, -y, z]$	[2]
3	$[y, -x, -z]$	[3]
4	$[-y, x, -z]$	[4]
5	$[-y, x, z]$	[5]
6	$[y, -x, z]$	[6]
7	$[-x, -y, -z]$	[7]
8	$[x, y, -z]$	[8]
9	$\left[x+\frac{1}{2}, y+\frac{1}{2}, z+\frac{1}{2}\right]$	[9]
10	$\left[\frac{1}{2}-x, \frac{1}{2}-y, z+\frac{1}{2}\right]$	[10]

continued ...

Table 9

No.	position	mapping
11	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[11]
12	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[12]
13	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[13]
14	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[14]
15	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[15]
16	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[16]